

Course Code : 2104CS352**Date** : 26-04-2023**Course Name** : Internet of Things**Duration** : 150 Minutes**Total Marks** : 70**Instructions:**

1. Attempt all the questions.
2. Figures to the right indicates maximum marks.
3. Make suitable assumptions wherever necessary.

Q.1 (A) Draw 7 layer architecture of IoT. **4****(B)** List down various protocols for Serial communication in IoT. **3****OR**

List down various protocols for Wireless communication protocols in IoT.

(C) List and Describe characteristics of IoT. **7****OR**

Compare Bluetooth protocol and Wi-Fi protocol.

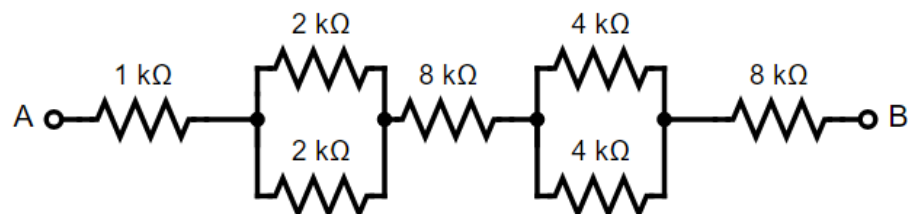
Q.2 (A) State the following Terms: **4**

- 1) Development board
- 2) Microprocessor
- 3) Microcontroller
- 4) SoC

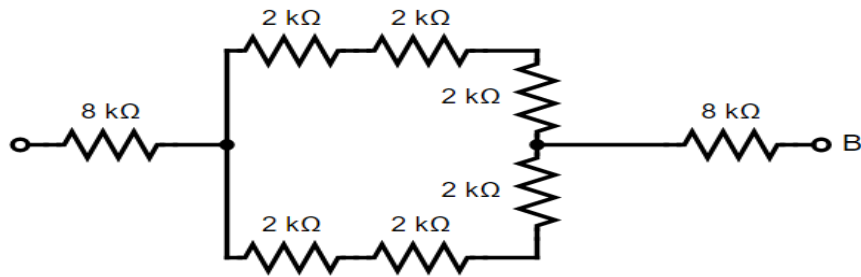
(B) Write down Kirchhoff's current law for node in circuit. **3****OR**

List down the SI unit of

- 1) Resistance
- 2) Inductance
- 3) Capacitance
- 4) Current
- 5) Voltage
- 6) Power

(C) Find equivalent Resistance between node A and B. **7****OR**

Find equivalent Resistance between node A and B.



Q.3 (A) Draw pin diagram and Layout of Relay. **4**

(B) Explain sensor and review the functionality of LDR. **3**

OR

Explain actuator and review the functionality Servo-motor.

(C) Write an Arduino code to measure the gas level in the room. **7**

OR

Write an Arduino code for the detect obstacle. Turn On the LED when obstacle detected, else keep LED turn OFF.

Q.4 (A) Demonstrate the functionality of void setup () and void loop () in detail with code structure. **4**

(B) Explain an Arduino IDE function to read and write digital value to Arduino pin. **3**

OR

Explain analogRead() and analogWrite() .

(C) Draw Arduino UNO Block diagram and list all the components on board. **7**

OR

Write an Arduino code to the control the brightness of LED (Pin 9) as per the Input Voltage from Potentiometer (Analog Pin A0). (Use map())

Q.5 (A) Explain various Services of Cloud computing in the IoT. **4**

(B) Write down benefits of Cloud Computing over Local computing. **3**

OR

Define common architecture of IoT application.

(C) Use the Blynk Library function to read the value of virtual pin V0 and pin V1 from the Blynk cloud. **7**

OR

Use the Blynk Library function to write HIGH or 1 to virtual pin V0 and LOW or 0 to virtual pin V1.
